COLLEGE ENGINEERING PROJECT · PROJECT 33 OF 55

REMAINING USEFUL LIFE PREDICTION OF AIRCRAFT COMPONENTS USING MACHINE LEARNING

Domain: Aerospace Prognostics & NASA C-MAPSS AI | **Module:** project_33_aircraft_rul

PROJECT ABSTRACT

Remaining Useful Life Prediction of Aircraft Components Using Machine Learning is a comprehensive and crucial project aimed at addressing unexpected equipment failures, costly downtime, and safety risks in modern aerospace environments. As aviation complexity accelerates, airlines and manufacturers increasingly rely on interconnected systems that continuously generate complex operational metrics, making them prime targets for sudden mechanical breakdowns if component degradation is not proactively identified. Traditional maintenance methods often rely on scheduled or reactive servicing, which can be time-consuming, resource-intensive, and fundamentally inefficient against rapidly developing, complex mechanical anomalies and subtle wear and tear. To address these persistent challenges and limitations, this study proposes an advanced, multi-layered approach combining cutting-edge deep learning techniques with intelligent machine learning algorithms for efficient, automated, and real-time aircraft health monitoring and remaining useful life (RUL) prediction. The system utilizes specialized deep learning architectures to continuously capture and extract key behavioral features from extensive flight sensor streams, such as fluctuating engine temperatures, vibration patterns, pressure variations, and operational speed deviations. Once these underlying structural patterns are mapped, the extracted features are fed into classification and forecasting algorithms designed to enhance decision-making, optimize computational overhead, improve prediction accuracy, and drastically reduce false alarm and misclassification rates. By profiling what constitutes normal, routine operation for each distinct aircraft component, the model establishes a dynamic baseline that adapts to changing flight load requirements over time. The proposed model is rigorously trained, validated, and tested on robust datasets encompassing diverse aerospace sensor logs, operational flight cycles, and simulated degradation scenarios. This enables the framework to accurately distinguish between healthy component operation and critical anomalies—such as turbine blade degradation, hydraulic system wear, and structural imbalances—with high precision and minimal latency. Experimental results and comprehensive performance evaluations demonstrate that the seamless integration of AI-driven feature extraction and machine learning models provides significantly improved classification and regression performance, resilience, and adaptability compared to traditional legacy maintenance approaches.

📌 Project Metadata & Specifications

Project Title	REMAINING USEFUL LIFE PREDICTION OF AIRCRAFT COMPONENTS USING MACHINE LEARNING
Domain / Specialization	Aerospace Prognostics & NASA C-MAPSS AI
Source Directory	`project_33_aircraft_rul`
Backend REST API	`http://localhost:5033` (Flask Python)
Frontend Interface	Interactive Dashboard (`frontend/index.html`)
Execution Mode	Hybrid Dual-Mode (Online API & Offline Autonomous Simulation)

🏗️ System Methodology & Implementation Phases

PROJECT 33 — PHASE 1

Phase 1 — Requirements, Data and Prototype

The first phase will convert the project concept into a measurable prototype for industrial predictive analytics/computer vision. The work begins by defining the exact input, output, user flow and evaluation criteria from the project abstract. A small requirements specification will identify the primary prediction/detection task, expected classes or target value, acceptable response time, and the form in which results must be displayed.

1. Problem Definition and Scope

Define the operational scenario, target users, system boundaries, assumptions, success criteria and non-functional requirements. Separate what the model must predict from what the application must display or recommend. Identify cases that should be rejected, flagged for review, or returned with low confidence.

2. Data Preparation

Identify a suitable dataset or approved collection procedure matching the abstract. Build a data dictionary, remove unusable records, handle missing values, normalize or standardize numerical fields where appropriate, and verify labels. For image/video/audio/text projects, define preprocessing, augmentation and quality-control rules. Split data into training, validation and test sets without leakage.

3. Baseline Model and Experiment Setup

Implement a reproducible baseline using the algorithm/model family stated in the abstract when one is specified. Establish a simple reference result before introducing additional complexity. Record dataset version, preprocessing settings, model configuration, random seed, training time and evaluation metrics so that later improvements can be compared objectively.

4. Initial Application Prototype

Build the minimum user-facing workflow needed to demonstrate the core task. The prototype should accept valid input, run preprocessing, invoke the baseline model, and return the predicted result with an understandable confidence/score where supported. Keep model logic separate from the interface so the model can be replaced without rewriting the application.

Phase 1 Deliverables

Requirements and scope note; prepared dataset or data pipeline; baseline model; experiment log; initial application/prototype; and a short Phase 1 validation report documenting what works, what remains uncertain, and the measurements to be improved in Phase 2.

Project-Specific Phase 1 Work Items
• Create a clean feature/data dictionary directly from the variables described in the abstract.
• Establish baseline metrics before optimization and preserve the baseline for comparison.
• Document assumptions and data limitations that could affect generalization.

PROJECT 33 — PHASE 2

Phase 2 — Optimization, Integration and Validation
The second phase will turn the Phase 1 prototype into a validated project for industrial predictive analytics/computer vision. The emphasis is on improving model reliability, connecting the complete workflow, testing realistic edge cases, and producing evidence that the final system meets the defined objectives.
1. Model Improvement
Compare relevant model configurations, tune important hyperparameters, address class imbalance where present, and evaluate whether additional features or augmentation improve generalization. Use validation data for tuning and reserve the test set for final evaluation. Prefer measurable improvements over unnecessary model complexity.
2. End-to-End Integration
Connect the data/input layer, preprocessing, inference engine, result-processing logic, database or file storage where required, and user interface. Add input validation, error handling, logging and clear status messages. If recommendations, alerts, summaries, risk scores or dashboards are part of the abstract, implement them as a separate decision-support layer around the model.
3. Testing and Evaluation
Perform functional, model, integration and usability tests. Evaluate accuracy or regression error as appropriate, along with precision, recall, F1, confusion matrix, ROC-AUC, latency or other task-specific measures where relevant. Test difficult, noisy, incomplete and out-of-distribution inputs and document failure cases instead of hiding them.
4. Demonstration and Final Evidence
Prepare representative test cases, screenshots or result tables, model-comparison evidence and a final demonstration workflow. Document limitations, data constraints, privacy/security considerations where applicable, and future extensions. The final report should distinguish measured results from proposed future work.
Phase 2 Deliverables
Optimized model; integrated application; test and evaluation report; representative evidence; limitations and risk register; deployment/demo instructions; and final project documentation suitable for academic review or demonstration.
Project-Specific Phase 2 Validation
• Compare the final model with the Phase 1 baseline using the same held-out evaluation set.
• Document representative success cases and failure cases instead of reporting only aggregate accuracy.
• Prepare a final demonstration script covering input, processing, output, validation and limitations.
Final Expected Outcome
A reproducible, tested prototype that implements the project concept described in the supplied abstract, together with measurable evaluation evidence and clear documentation of limitations, assumptions and future enhancement opportunities.

🚀 Local Deployment & Execution Guide

Follow the steps below to initialize and demonstrate the system:

1. Backend Server Setup

```
# Navigate to the backend directory cd "/Users/balamurugan/Documents/orbyte globel
/college project/project_33_aircraft_rul/backend" # Install project dependencies pip
install -r requirements.txt # Run the Flask API service python3 app.py
```

The backend microservice will listen for requests at `http://localhost:5033`.

2. Frontend User Interface

- Open `/Users/balamurugan/Documents/orbyte globel /college project/project_33_aircraft_rul/frontend/index.html` directly in Google Chrome, Firefox, or Safari.
- Alternatively, launch via the central **Master Launchpad Portal (`index.html`)** located at the root directory.
- The user interface is designed with **hybrid dual-mode resilience**: when the backend API is live, real-time inferences are queried; when offline, authentic client-side mathematical models simulate sensor telematics.